



Association between disinfectant resistance genes in *exoU* and *exoS* *Pseudomonas aeruginosa* with sensitivity to multipurpose disinfecting solutions and antibiotics

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ABSTRACT

Background: The aim of this study was to determine differences in susceptibility of *exoU* and *exoS* *Pseudomonas aeruginosa* to multi-purpose disinfecting solutions (MPDS) and antibiotics and their association with disinfectant resistance genes.

Methods: A total of 16 *exoU* and 24 *exoS* *P. aeruginosa* keratitis isolates were tested for susceptibility against four multipurpose disinfecting solutions (OPTI-FREE PureMoist, Acuvue RevitaLens, Biotrue, and Renu Advanced Formula) using minimum inhibitory (MIC) and minimum bactericidal (MBC) concentrations. Antibiotic susceptibility to ciprofloxacin, levofloxacin, gentamicin, and tobramycin was determined by MIC. Whole genome sequences from 27 isolates were utilized to identify disinfectant resistance genes and single nucleotide polymorphisms (SNPs) in the genes using the Comprehensive Antibiotic Resistance Database (CARD) and Geneious Prime respectively. Polymerase chain reaction (PCR) was used to detect the disinfectant resistance genes *qacE*, *qacEΔ1*, and *sugE1*. Correlation between disinfectant resistance genes and their SNPs with disinfectant and antibiotic MICs was determined by Spearman's Rho.

Results: All MPDS were bactericidal at full strength, but upon dilution OPTI-FREE Puremoist and Renu Advanced Formula were the most active against the *exoU* and *exoS* groups correspondingly. *ExoU* showed significantly higher resistance than *exoS* ($p \leq 0.04$) to levofloxacin (37.5 % vs. 8.3 %), gentamicin (43.8 % vs. 0 %), and tobramycin (37.5 % vs. 0 %). The *qacEΔ1* and *sugE1* genes along with several SNPs in the resistance genes were more common in the *exoU* group. SNPs in disinfectant resistance genes correlated with MPDS MICs/MBCs, notably *mexQ* Arg1036His ($R \geq -0.38$, $p \leq 0.04$), Gly505Asp ($R \geq 0.40$, $p \leq 0.04$) and *opmD* Gly269Ser ($R \geq 0.43$, $p \leq 0.03$). *QacEΔ1* and *sugE1* were linked to increased MIC ($R \geq 0.54$, $p < 0.01$) and MBC ($R \geq 0.50$, $p < 0.01$) for Renu Advanced Formula. Also, multiple SNPs and *qacEΔ1* and *sugE1* showed moderate to large positive associations with antibiotic resistance.

Conclusions: *ExoU* strains had distinct MPDS susceptibility patterns, with higher disinfectant resistance genes prevalence and mutations, contributing to MPDS and antibiotics resistance.

1. Introduction

The more than 140 million people across the world who wear contact lens [1,2] derive visual and vocational benefits. However, contact lens wear is a risk factor for developing microbial keratitis (MK) or corneal infection; see Supplementary Table S1 for full list of abbreviations) if used overnight or if contact lens hygiene is poor [3]. Whilst MK is a rare, affecting approximately 5 out of every 10,000 people who wear contact

lenses [4], if not appropriately therapeutically managed, it can lead to perforation of the cornea and blindness [5]. *Pseudomonas aeruginosa* is responsible for ≥ 66 % of all contact lens-related keratitis cases [6–8].

One of the most likely sources of microbes causing MK is those attached to a contact lens. To reduce the number of bacteria contaminating reusable contact lenses, multipurpose disinfecting solutions (MPDS) are used that contain disinfectants such as quaternary ammonium compounds (QACs) and biguanides in varying formulations and

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concentrations [9,10]. However, improper use of MPDS, such as topping up or reuse of solutions, can expose bacteria to sub-lethal disinfectant levels which allow the bacteria to survive and may help them to become resistant to the MPDS [11]. In *P. aeruginosa*, disinfectant resistance mechanisms include the action of small multidrug resistance (SMR) proteins, which are embedded in the inner cytoplasmic membrane. SMR family proteins function as proton-dependent efflux pumps [12]. These SMR proteins can be classified into three subgroups, SUG (suppressor of *groEL* mutation proteins), SMP (small multidrug pumps), and PSMR (paired small multidrug resistance proteins) [12].

The SMP group genes *qacE* and *qacEΔ1* (a mutated form of *qacE*) are often found on plasmids and integrons facilitating resistance transmission [13]. *Qac* genes confer resistance to QACs and other cationic compounds, including biguanides, diamidines, and intercalating dyes [14]. *Qac* genes are often located near antibiotic resistance genes on integrons, indicating that they may be co-transferred between bacterial populations. This suggests that exposure to disinfectants could play a role in the development of antibiotic resistance.

In addition, according to the Comprehensive Antibiotic Resistance Database (CARD), a number of efflux pumps *mexGHI-OpmD*, *mexJKI-OprM*, *mexPQ-OprE*, *mexVW-OprM*, *mexXY-OprM*, *triABC-OpmH* and their regulatory genes (*mexL*, *mexZ*, *parR*, *parS*, *pmpM*, *soxR*, *yajC*) are also considered to play an important role in disinfectant resistance [15]. In *P. aeruginosa*, efflux pumps prevent the intracellular accumulation of toxic substances, such as disinfectants and antibiotics, enhancing pathogen survival. Overexpression of these pumps is associated with multidrug resistance [16–18].

There are two lineages of *P. aeruginosa*, based on the presence of *exoU* or *exoS* genes. *P. aeruginosa* releases the exotoxins encoded by these genes after adhering to the corneal cell surface [19,20]. Strains possessing *exoU* use the gene to produce a powerful phospholipase that causes acute cell death [5,21]. Strains possessing *exoS* invade and multiply inside mammalian cells, with the product of the gene having guanosine triphosphatase (GTPase) and adenosine diphosphate ribosyl transferase (ADPRT) activity [21–24]; the host cells then die through apoptosis. Strains carrying *exoU* cause more severe infection in animals and humans [25–27]. Previous studies have indicated that strains carrying *exoU* are linked to higher resistance to antibiotics [28,29]. Moreover, studies have demonstrated a co-selection relationship between the *exoU* gene and antimicrobial resistance, including resistance to disinfectants, in *exoU*-positive strains from both ocular and non-ocular origins [29–32].

The aim of the current study was to examine how these two lineages of *P. aeruginosa* respond to MPDS, how they differ in terms of disinfectant resistance genes and whether there was any association between disinfectant resistance genes and single nucleotide polymorphisms (SNPs) in disinfectant resistance genes with resistance to MPDS or antibiotics.

2. Methods

2.1. Bacterial isolates

16 *exoU* and 24 *exoS* *P. Aeruginosa* isolates (Supplementary Table S2) from MK collected from Australia and India were revived from the microbial culture collection of the School of Optometry and Vision Science, UNSW Sydney, NSW, Australia. Nutrient agar (Oxoid Ltd., Basingstoke, Hampshire, UK) was used to revive the bacteria

2.2. Susceptibility of *exoU* and *exoS* *P. aeruginosa* to multipurpose disinfecting solutions

The susceptibility of bacterial strains to four commercially available multipurpose disinfecting solutions (MPDS) for contact lenses, Acuvue RevitaLens (Johnson & Johnson Vision, Jacksonville, Florida, USA), Renu Advanced Formula (Bausch and Lomb, Rochester, NY, USA)

Biotrue (Bausch and Lomb) and OPTI-FREE PureMoist (Alcon, Fort Worth, TX, USA) (Table 1), was conducted using a modified micro-dilution broth method, based on the Clinical and Laboratory Standards Institute (CLSI) guidelines [33] and previously described protocols [34,35]. Each MPDS was prepared as a series of twofold dilutions using sterile phosphate buffered saline to produce concentrations ranging from 50 % (1:2) to 3.12 % (1:32). Bacterial suspensions were prepared in Mueller-Hinton broth, adjusted to 1×10^8 CFU/mL ($OD_{660} = 0.1$), and further diluted to achieve a final inoculum of 1×10^5 CFU/mL in each well by adding 20 μ L bacterial suspension into 200 μ L of the serially diluted MPDS into the wells of 96-well microtiter plates (Costar, Corning, NY, USA). The microtiter plates were then incubated at 37 °C for 18–24 h and turbidity were measured at 660 nm using a spectrophotometer (FLUOstar Omega, BMG LABTECH, Germany) to determine the MIC (defined as the lowest concentration of MPDS that inhibited visible bacterial growth). MBC values were obtained by plating 10 μ L samples from wells corresponding to the MIC and the two lower concentrations onto Dey Engley agar (Sigma Aldrich), followed by incubation at 37 °C for 18–24 h and counting the number of colony forming unit (CFU). The MBC was identified as the lowest MPDS concentration achieving a ≥ 99.9 % (3 log units) reduction in bacterial viability. All assays were conducted in triplicate and repeated twice for accuracy and reproducibility.

2.3. Antibiotic susceptibility testing

To measure the MIC for fluoroquinolone (ciprofloxacin, levofloxacin), and aminoglycoside (gentamicin, tobramycin) antibiotics, each antibiotic was serially diluted in sterile 96-well plates with the Mueller-Hinton broth (cation-adjusted, Becton Dickinson and Company, Franklin Lakes, NJ, USA) and the antibiotics concentrations ranged from 5120 μ g/mL to 0.25 μ g/mL. Overnight grown bacterial culture was diluted with fresh media to achieve a final concentration of 1×10^5 CFU/mL. 100 μ L of diluted antibiotic and 100 μ L of diluted bacteria were added in each well and incubated at 37 °C for 18–24 h. The MIC was determined as the lowest antibiotic concentration that inhibited bacterial growth. Susceptibility levels were interpreted following the guidelines from the European Committee on Antimicrobial Susceptibility Testing (EUCAST) [36] and the Clinical and Laboratory Standards Institute (CLSI) [37]. Both intermediate resistance and full resistance to antibiotics were categorized as resistant. The antibiotics resistance of all the strains included in current study have been published previously [28,38].

2.4. Detection of disinfectant resistance genes

Whole genome sequences (WGS) for 27 of the 40 isolates (14 *exoU* and 13 *exoS*) were available from NCBI, with BioProject numbers PRJNA590804 and PRJNA431326 [39,40]. The CARD database was used to identify disinfectant resistance genes with ≥ 95 % identity in these 27 isolates [15]. Additionally, the WGS data were employed in BLAST searches to retrieve sequences for the CARD identified disinfectant resistance genes (*mexG*, *mexH*, *mexI*, *mexJ*, *mexK*, *mexL*, *mexP*, *mexQ*, *mexV*, *mexW*, *mexX*, *mexY*, *mexZ*, *opmD*, *opmE*, *opmH*, *oprM*, *pmpM*, *triA*, *triB*, *triC*, *soxR*, *yajC*, *parR*, and *parS*). These gene sequences were then aligned against the reference sequence of *P. aeruginosa* PAO1 for *exoS* strains and *P. aeruginosa* PA14 for *exoU* strains to detect SNPs using Geneious Prime version 2024.0.7 [41]. The SIFT (Sorting Intolerant from Tolerant) algorithm was utilized to assess potential impact of the identified novel mutations on protein function [42].

The CARD database did not identify the genes *qacE*, *qacEΔ1* and *sugE* in any isolate with identities above 50 %, yet these have previously been found by BLAST searches in some of the isolates used in the current study [43]. Therefore, polymerase chain reaction (PCR) was used to identify if the strains in the current study possessed these disinfectant resistance genes. Isolates were inoculated into Tryptic soya broth (TSB;

Table 1

Composition of multipurpose disinfectant solutions (MPDS).

MPDS	Manufacturer	Disinfectants	Surfactant and other ingredients
Acuvue RevitaLens	Johnson and Johnson Vision, Jacksonville, FL, USA	Alexidine dihydrochloride 1.6 ppm, polyquaternium-1 3 ppm	Tetronic 904, sodium citrate, sodium chloride, boric acid, sodium borate decahydrate, ethylenediaminetriacetic acid.
Renu Advanced Formula	Bausch and Lomb, Rochester, NY, USA	Polyaminopropyl biguanide (PAPB) (Polyhexamethylene biguanide – PHMB) 0.5 ppm, polyquaternium-1 1.5 ppm, alexidine 2 ppm	Poloxamine, poloxamer 181, sodium chloride, boric acid, sodium borate, ethylenediaminetriacetic acid, diglycine.
Biotrue	Bausch and Lomb, Bridgewater, NJ, USA	Polyaminopropyl biguanide (PAPB) (Polyhexamethylene biguanide – PHMB) 1.3 ppm, polyquaternium-1 1 ppm	Poloxamine, sulfobetaine, sodium chloride sodium borate, boric acid, ethylenediaminetriacetic acid, hyaluronan.
Opti-Free PureMoist	Alcon, Fort Worth, TX, USA	Polyquaternium-1 10 ppm, Aldox 6 ppm	Tectronic 1304, polyoxyethylene-polyoxybutylene copolymer, sodium citrate, sodium chloride, boric acid, aminomethyl-propanol, sorbitol, ethylenediaminetriacetic acid.

Oxoid Ltd., Basingstoke, UK) and incubated at 37 °C for 18–24 h. DNeasy Blood and Tissue Kit (Qiagen, Hilden, Germany) was used to extract DNA from fresh culture following the manufacturer's guidelines. The quantity and purity of the extracted DNA was measured with Nanodrop (ND-1000, Thermo Fisher Scientific, Waltham, MA Nanodrop). After extraction, DNA was preserved at –20 °C for PCR. The presence of *qacE*, *qacEΔ1* and *sugE1* was detected by PCR using gene specific primers following cycling conditions described previously [44]. In Table 2 and Supplementary Table S3 and S4, the primer sequences, PCR cycling condition, and components of PCR reaction mixture are presented. Following PCR, the amplified DNA fragments were visualized on 1.0 % agarose gel with Sybr Safe (Thermo Fisher Scientific, USA) as DNA staining dye to confirm the size of the fragments.

2.5. Statistical analysis

To find out differences between susceptibility to MPDS and antibiotics and possession of disinfectant resistance genes and SNPs in the disinfectant resistance genes of *exoU* and *exoS* groups, Fisher's exact test was performed using GraphPad prism, 2024, v8.0.2.26. Correlations were studied using Spearman's Rho. In all analysis, a p-value of less than 0.05 was considered as statistically significant.

3. Results

3.1. Susceptibility of 16 *exoU* and 24 *exoS* *P. Aeruginosa* to MPDS

The results of MIC and MBC of 16 *exoU* and 24 *exoS* *P. aeruginosa* isolates tested against four commonly available MPDS are summarized in Table 3 (and Supplementary Table S5 for individual strains). The MPDS susceptibility results of several of the strains included in current study have been published previously [38,43], and corresponded to those previous data. For the *exoU* group, OPTI-FREE Puremoist (median MIC = 12.5 %, median MBC = 25 %) was the most effective with Biotrue being the least effective (median MIC = 25 %, median MBC = 50 %). For the *exoS* group, Renu Advanced Formula (MIC 6.25 % and MBC 12.5 %) was the most effective and Biotrue was the least effective (median MIC = 25 %, median MBC = 50 %).

Table 2

Sequence of primers used in this study.

Gene	5' to 3' primer sequence	Product size
<i>qacE</i>	F- TGCCTTCCTGGATCTATCTG R- GACGATGCCAATGCCTTC	206 bp [44]
<i>qacEΔ1</i>	F- TTGTTATCGCAATAGTTG R- AATGGCTGTAATTATGAC	202 bp [44]
<i>sugE1</i>	F- TTGGAGTGGGGTCGCAGTA R- CACGCCAAGGTATCGCATC	187 bp This study

Table 3MIC and MBC of 16 *exoU* and 24 *exoS* *P. aeruginosa* to four different MPDS.

Strain type	Median MIC and MBC for each MPDS							
	OPTI-FREE Puremoist		Renu Advanced Formula		Biotrue		Acuvue RevitaLens	
	MIC (%)	MBC (%)	MIC (%)	MBC (%)	MIC (%)	MBC (%)	MIC (%)	MBC (%)
<i>exoU</i>	12.5	25	18.75	37.5	25	50	12.5	50
<i>exoS</i>	12.5	25	6.25	12.5	25	50	12.5	37.5

MIC = minimum inhibitory concentration; MBC = minimum bactericidal concentration

3.2. Susceptibility of 16 *exoU* and 24 *exoS* *P. Aeruginosa* to antibiotics

A greater proportion of *exoU* isolates were resistant to the four tested antibiotics than *exoS* isolates (Table 4), and apart from ciprofloxacin, these differences reached statistical significance ($p \leq 0.04$; Table 4).

3.3. Differences in SNPs in disinfectant resistance genes of 14 *exoU* and 13 *exoS* *P. Aeruginosa*

As all 27 strains (14 *exoU* and 13 *exoS*) possessed the CARD identified disinfectant associated genes, the differences in SNPs of these genes were evaluated. All the SNPs, except in the *parR* and *parS* genes, detected in the current study are novel. A comparative analysis of SNPs in disinfectant resistance genes revealed a distinct genetic profile between *exoU* and *exoS* *P. aeruginosa* isolates. Notably, multiple SNPs in disinfectant resistance genes (*mexG*, *mexH*, *mexI*, *mexJ*, *mexK*, *mexL*, *mexQ*, *mexV*, *mexW*, *mexX*, *mexY*, *mexZ*, *opmD*, *pmpM*, *triA*, *parR*) were significantly enriched in *exoU* isolates (Table 5, Supplementary Table S6). Mutations in *mexG* (Ala58Thr), *mexJ* (Phe13Leu), *mexQ* (Pro172Leu), *pmpM* (Phe422Ile, Glu470Ala), *mexW* (Gly932Glu), *opmD* (Ala85Thr), *opmE* (Pro59Ser), *triA* (Val304Asp, Gln316Glu, Gly317Ser, Ala319Val), *parR* (Leu153Arg, Ser170Asn), *mexX* (Ala30Thr), *mexY* (Ile536Val, Gly589Ala), and *mexZ* (Gly89Ser, novel functional SNP) were detected only in *exoU* strains (ranging from 35 % to 92.9 %, $p \leq 0.04$; Table 5). Conversely, for *opmE* (Ser46Gly) 76.9 % of *exoS* strains carried this mutation compared to only 7.1 % in *exoU* strains ($p < 0.01$) (Table 5).

Mutations in *mexI* (Ala782Glu), *mexL* (Ser4Pro), *mexQ* (Gly505Asp) and *triA* (Gly34Ser) were highly prevalent in *exoU* strains, with 92.9 %

Table 4Antibiotic resistance of 16 *exoU* and 24 *exoS* *P. aeruginosa* against ciprofloxacin, levofloxacin, gentamicin and tobramycin.

Antibiotics	Resistant <i>exoU</i> (%)	Resistant <i>exoS</i> (%)	P value
Ciprofloxacin	50.0	41.7	>0.05
Levofloxacin	37.5	8.3	0.04
Gentamicin	43.8	0	<0.01
Tobramycin	37.5	0	<0.01

Table 5

Significantly different SNPs in disinfectant resistance genes of 14 *exoU* and 13 *exoS* *P. aeruginosa*.

Gene*	SNPs	Frequency of possession (%) <i>exoU:exoS</i>	Difference between <i>exoU</i> and <i>exoS</i> (p-value)
<i>mexG</i>	Ala58Thr	43:0	0.02
<i>mexH</i>	Asp302Glu	57:8	0.01
<i>mexI</i>	Ala782Glu	93:15	<0.01
<i>mexJ</i>	Phe13Leu	93:0	<0.01
	Ala314Pro	79:23	<0.01
<i>mexK</i>	Lys694Arg	79:8	<0.01
<i>mexL</i>	Ser4Pro	93:31	<0.01
<i>mexQ</i>	Pro172Leu	36:0	0.04
	Gly505Asp	93:46	0.01
<i>mexV</i>	Ala229Gly	100:69	0.04
<i>mexW</i>	Gly932Glu	43:0	0.02
<i>mexX</i>	Ala30Thr	64:0	<0.01
<i>mexY</i>	Ile536Val	57:0	<0.01
	Gly589Ala	35:0	<0.01
	Gln840Glu	64:15	0.02
	Asn1036Thr	57:8	0.01
<i>mexZ</i>	Gly89Ser	35:0	<0.01
<i>opmD</i>	Ala85Thr	36:0	0.04
	Gly269Ser	57:8	0.01
	Ala270Gly	64:15	0.02
<i>opmE</i>	Ser46Gly	7:77	<0.01
	Pro59Ser	36:0	0.04
<i>parR</i>	Leu153Arg	71:0	<0.01
	Ser170Asn	79:0	<0.01
<i>pmpM</i>	Phe422Ile	54:0	<0.01
	Glu470Ala	71:0	<0.01
<i>triA</i>	Gly34Ser	93:23	<0.01
	Val304Asp	43:0	0.02
	Gln316Glu	57:0	<0.01
	Gly317Ser	71:0	<0.01
	Ala319Val	43:0	0.02

*No strain had mutations in *soxR* and *yajC*. All mutations are new except mutations in *parR*, **Bold** indicates presumed functional SNP.

of *exoU* strains carrying these mutations, respectively, compared to much lower frequencies in *exoS* strains (15.4 %, 30.8 %, 46.2 %, and 23.1 % respectively; $p \leq 0.01$) (Table 5). Mutations present in 78.6 % of *exoU* strains were *mexJ* (Ala314Pro), *mexK* (Lys694Arg), and *parR* (Ser170Asn), with significant differences compared to *exoS* strains (0 % to 23.1 %; $p < 0.01$). Mutations in *mexH* (Asp302Glu), *mexY* (Ile536Val), and *opmD* (Gly269Ser) were at 57.2 % in *exoU* strains and 7.7 % in *exoS* group ($p = 0.01$). Moreover, 100 % *exoU* strains had amino acid change in the Ala229Gly in *mexV* gene while 69.2 % *exoS* strains carried this SNP. No amino acid changes were found in *soxR* and *yajC* genes and SNPs that were not significantly different between *exoU* and *exoS* are presented in Supplementary Table S7.

3.4. Distribution of SMR disinfectant resistance genes in 16 *exoU* and 24 *exoS* *P. aeruginosa*

The frequency of SMR disinfectant resistant genes is presented in Table 6 and Supplementary Table S8. All *exoU* strains (100 %) and all except one *exoS* (95.8 %) isolates possessed *qacE*. The possession of *sugE1* and *qacEΔ1* varied significantly ($p < 0.01$) between *exoU* and *exoS* groups, with *exoU* isolates more commonly possessing these genes (43.8 % vs. 4.2 % and 37.5 % vs. 0 % respectively).

Table 6

Possession of SMR disinfectant resistance genes by the 16 *exoU* and 24 *exoS* *P. aeruginosa*.

Gene	<i>exoU</i> (16)	<i>exoS</i> (24)	P value
<i>qacE</i>	100 %	95.8 %	>0.05
<i>qacEΔ1</i>	37.5 %	0 %	<0.01
<i>sugE1</i>	43.8 %	4.2 %	<0.01

3.5. Association of SNPs in disinfectant resistance genes and MICs and MBCs to MPDS between 14 *exoU* and 13 *exoS* *P. aeruginosa*

SNPs in several disinfectant resistance genes were associated with relative resistance to MPDS (Table 7). The SNP in *mexQ* resulting in the amino acid change Arg1036His, was significantly negatively associated with levels of MIC or MBC for all four MPDS ($R \geq -0.38$; $p \leq 0.04$). In other words, the possession of this SNP was associated with lower MICs or MBCs, and the presence of the SNP explained between 15 to 32 % of the variance in MIC or MBC data. Another SNP, Gly505Asp in *mexQ*, was significantly positively associated ($R \geq 0.40$; $p \leq 0.04$) with the MICs and MBCs of all MPDS, except the MIC for Acuvue RevitaLens ($p = 0.06$). Similarly, a SNP in *opmD*, Gly269Ser, was significantly positively associated ($R \geq 0.43$; $p \leq 0.03$), with MICs and MBCs, except the MIC for Acuvue RevitaLens ($p = 0.08$).

3.6. Association of possession of *qacE*, *qacEΔ1*, *sugE1* and MICs and MBCs to MPDS between 16 *exoU* and 24 *exoS* *P. aeruginosa*

The possession of *qacE* was not associated with MICs or MBCs to MPDS (Table 8). Possession of *qacEΔ1* was significantly positively associated with Renu Advanced Formula MIC ($R = 0.54$, $p < 0.01$) and MBC ($R = 0.50$, $p < 0.01$), Biotrue MIC ($R = 0.54$, $p < 0.01$), and Acuvue RevitaLens MIC ($R = 0.42$, $p = 0.02$) and MBC ($R = 0.38$, $p = 0.04$; Table 8). Possession of *sugE1* was significantly positively associated with Renu Advanced Formula MIC ($R = 0.57$, $p < 0.01$) and MBC ($R = 0.57$, $p < 0.01$), and Biotrue MIC ($R = 0.43$, $p = 0.02$; Table 8).

3.7. Association between SNPs present in disinfectant resistance genes and antibiotic resistance of 14 *exoU* and 13 *exoS* *P. aeruginosa*

SNPs in *triA* (Val304Asp, Ala319Val) showed moderate associations with ciprofloxacin resistance, ($R = 0.41$ and $p = 0.03$; Table 9). Except for these mutations in *triA* gene, mutations in all other disinfectant resistance genes had no association with the resistance to ciprofloxacin (Table 9). For levofloxacin resistance, mutations in *mexG* (Ala58Thr), *mexQ* (Pro172Leu), *mexW* (Gly932Glu), *opmD* (Ala85Thr), *triA* (Val304Asp, Gln316Glu, Ala319Val), *mexX* (Ala30Thr), and *mexZ* (Gly89Ser) demonstrated strong associations ($R = 0.67$ and $p < 0.01$) while mutations leading to amino acid changes in *mexJ* (Phe13Leu), *pmpM* (Phe422Ile), *opmE* (Pro59Ser), *parR* (Leu153Arg), *mexY* (Gly589Ala) showed moderate associations, with R ranging from 0.42 to 0.48, and $p \leq 0.03$. In case of aminoglycoside resistance, SNPs in *mexQ* (Gly505Asp) and *mexV* (Ala229Gly) had no association with either gentamicin or tobramycin but a mutation in *ompE* (Ser46Gly) was negatively associated with aminoglycosides resistances. Except these SNPs, mutations in all other genes showed a moderate to strong positive correlation with aminoglycosides. For instances, *mexL* (Ser4Pro), and *triA* (Gly34Ser) had moderate correlation ($R = 0.45$, $p = 0.02$) with both gentamicin and tobramycin and SNPs in *mexG* (Ala58Thr), *mexH* (Asp302Glu), *mexI* (Ala782Glu), *mexJ* (Phe13Leu), *mexQ* (Pro172Leu), *pmpM* (Phe422Ile, Glu470Ala), *mexW* (Gly932Glu), *opmD* (Ala85Thr, Gly269Ser, Ala270Gly), *opmE* (Pro59Ser), *triA* (Val304Asp, Gln316Glu, Ala319Val), *parR* (Ser170Asn), *mexX* (Ala30Thr), *mexY* (Ile536Val, Gly589Ala, Gln840Glu, Asn1036Thr), and *mexZ* (Gly89Ser) had large positive associations with aminoglycosides resistances.

3.8. Association between SMR disinfectant resistance genes and antibiotics resistance of 16 *exoU* and 24 *exoS* *P. aeruginosa*

As almost all the isolates possessed the *qacE* gene, so there was no relevant association between this gene and antibiotics resistance (Table 10). However, the *qacEΔ1* was exclusively found in *exoU* group showing a moderate correlation with ciprofloxacin resistance ($R = 0.46$, $p < 0.01$) and a stronger correlation with levofloxacin resistance ($R = 0.67$, $p < 0.01$). Possession of *qacEΔ1* was strongly correlated with

Table 7
Correlations between SNPs in disinfectant resistance genes and MICs and MBCs to MPDS between 14 *exoU* and 13 *exoS* keratitis isolates.

Gene	Amino acid change	OPTI-FREE Puremoist		MBC		Renu Advanced Formula		MBC		Biotrue		MBC		Acuvue RevitaLens		MBC	
		MIC R	P value	R	P value	MIC R	P value	R	P value	MIC R	P value	R	P value	MIC R	P value	R	P value
<i>mexG</i>	Ala58Thr	0.12	0.56	0.43	0.02	0.43	0.02	0.36	0.07	0.54	<0.01	0.26	0.18	0.19	0.35	0.29	0.14
<i>mexH</i>	Asp302Glu	0.23	0.27	0.47	0.01	0.30	0.13	0.23	0.24	0.40	0.04	0.14	0.49	0.29	0.14	0.25	0.21
<i>mexJ</i>	Ala314Pro	0.21	0.30	0.43	0.03	0.25	0.20	0.25	0.20	0.35	0.07	0.29	0.14	0.15	0.46	0.32	0.10
<i>mexL</i>	Ser4Pro	0.32	0.10	0.43	0.02	0.53	<0.01	0.46	0.01	0.36	0.06	0.39	0.04	0.20	0.32	0.37	0.06
<i>mexP</i>	Ala297Glu	0.31	0.12	0.31	0.11	0.37	0.05	0.31	0.12	0.52	<0.01	0.45	0.02	0.34	0.08	0.46	0.02
<i>mexQ</i>	Arg1036His	−0.49	<0.01	−0.57	<0.01	−0.38	0.04	−0.38	0.04	−0.55	<0.01	−0.55	<0.01	−0.40	0.04	−0.46	0.02
	Asp602Glu	0.21	0.30	0.35	0.08	0.22	0.27	0.22	0.27	0.55	<0.01	0.27	0.17	0.20	0.32	0.25	0.22
	Gly505Asp	0.47	0.01	0.60	<0.01	0.48	0.01	0.41	0.03	0.40	0.04	0.41	0.03	0.37	0.06	0.44	0.02
	Pro172Leu	0.26	0.18	0.38	0.04	0.52	<0.01	0.44	0.02	0.62	<0.01	0.32	0.10	0.37	0.06	0.36	0.07
<i>mexW</i>	Gly932Glu	0.23	0.21	0.40	0.04	0.39	0.04	0.31	0.11	0.47	0.01	0.19	0.33	0.27	0.17	0.24	0.23
<i>mexY</i>	Asn1036Thr	0.11	0.59	0.30	0.12	0.38	0.04	0.38	0.04	0.40	0.04	0.25	0.22	0.10	0.62	0.19	0.35
	Gly589Ala	0.23	0.24	0.42	0.03	0.43	0.02	0.36	0.07	0.54	<0.01	0.26	0.18	0.32	0.10	0.29	0.14
	Ile536Val	0.11	0.59	0.30	0.12	0.23	0.24	0.16	0.41	0.40	0.04	0.14	0.49	0.17	0.41	0.11	0.60
<i>mexZ</i>	Gly89Ser	0.26	0.18	0.38	0.04	0.52	<0.01	0.44	0.02	0.62	<0.01	0.32	0.10	0.37	0.06	0.36	0.07
<i>opmD</i>	Ala270Gly	0.21	0.29	0.36	0.04	0.34	0.09	0.34	0.09	0.47	0.01	0.32	0.11	0.15	0.44	0.21	0.28
	Ala85Thr	0.26	0.18	0.38	0.04	0.52	<0.01	0.44	0.02	0.62	<0.01	0.32	0.10	0.37	0.06	0.36	0.07
	Gly269Ser	0.43	0.03	0.49	<0.01	0.56	<0.01	0.56	<0.01	0.58	<0.01	0.52	<0.01	0.35	0.08	0.44	0.02
<i>opmE</i>	Pro59Ser	0.13	0.51	0.23	0.25	0.44	0.02	0.44	0.02	0.36	0.07	0.19	0.33	0.22	0.28	0.36	0.07
	Ser46Gly	−0.21	0.29	−0.39	0.04	−0.47	0.01	−0.53	<0.01	−0.46	0.01	−0.29	0.15	−0.14	0.49	−0.31	0.12
<i>opmH</i>	Ala471Ser	−0.29	0.14	−0.24	0.24	−0.52	<0.01	−0.52	<0.01	−0.44	0.02	−0.45	0.02	−0.18	0.36	−0.25	0.21
<i>parR</i>	Leu153Arg	0.11	0.60	0.33	0.09	0.28	0.15	0.28	0.15	0.38	0.04	0.23	0.25	−0.03	0.89	0.16	0.42
	Met59Ile	0.26	0.18	0.38	0.04	0.52	<0.01	0.44	0.02	0.62	<0.01	0.32	0.10	0.37	0.06	0.36	0.07
<i>parS</i>	Ser277Asn	−0.19	0.33	−0.16	0.43	−0.40	0.04	−0.40	0.04	−0.02	0.92	−0.04	0.85	−0.01	0.96	−0.04	0.83
<i>pmpM</i>	Glu470Ala	0.11	0.60	0.33	0.09	0.21	0.29	0.14	0.47	0.38	0.04	0.13	0.53	0.10	0.63	0.02	0.91
	Phe422Ile	0.23	0.24	0.42	0.03	0.43	0.02	0.36	0.07	0.54	<0.01	0.26	0.18	0.32	0.10	0.29	0.14
<i>triA</i>	Ala319Val	0.1	0.60	0.42	0.03	0.51	<0.01	0.43	0.02	0.54	<0.01	0.26	0.18	0.32	0.10	0.44	0.02
	Gln316Glu	0.22	0.27	0.47	0.01	0.44	0.02	0.44	0.02	0.40	0.04	0.25	0.22	0.17	0.41	0.39	0.04
	Glu291Ala	0.11	0.58	0.27	0.17	0.40	0.04	0.33	0.10	0.42	0.03	0.15	0.44	0.17	0.39	0.28	0.16
	Gly34Ser	0.11	0.60	0.27	0.17	0.39	0.04	0.33	0.10	0.24	0.24	0.16	0.42	−0.11	0.57	0.03	0.86
	Val304Asp	0.23	0.24	0.42	0.03	0.51	<0.01	0.43	0.02	0.54	<0.01	0.26	0.18	0.32	0.10	0.44	0.02
<i>triC</i>	Arg823Lys	0.35	0.07	0.42	0.03	0.43	0.02	0.36	0.07	0.44	0.02	0.29	0.15	0.32	0.10	0.29	0.14

MIC = minimum inhibitory concentration; MBC = minimum bactericidal concentration; R = Spearman's Rho correlation coefficient, **bold** numbers indicate $p < 0.05$.

Table 8Association of possession of *qacE*, *qacEΔ1*, *sugE1* and MICs and MBCs to MPDS between 16 *exoU* and 24 *exoS* keratitis isolates.

Gene	OPTI-FREE Puremoist				Renu Advanced Formula				Biotrue				Acuvue RevitaLens			
	MIC	P value	MBC	P value	MIC	P value	MBC	P value	MIC	P value	MBC	P value	MIC	P value	MBC	P value
<i>qacE</i>	−0.02	0.91	−0.08	0.69	0.14	0.47	0.14	0.47	−0.04	0.81	−0.04	0.85	−0.16	0.4	−0.15	0.45
<i>qacEΔ1</i>	0.16	0.39	0.31	0.11	0.54	<0.01	0.50	<0.01	0.54	<0.01	0.35	0.07	0.42	0.02	0.38	0.04
<i>sugE1</i>	−0.01	0.96	0.21	0.28	0.58	<0.01	0.57	<0.01	0.43	0.02	0.33	0.08	0.19	0.31	0.35	0.06

MIC = minimum inhibitory concentration; MBC = minimum bactericidal concentration; R = Spearman's Rho correlation coefficient, **bold** numbers indicate $p < 0.05$.**Table 9**Differences of SNPs and correlations between SNPs in disinfectant resistance genes and resistance to fluoroquinolones and aminoglycosides between 14 *exoU* and 13 *exoS* keratitis isolates.

Gene	SNPs	Correlation between SNP and resistance to ciprofloxacin (Spearman's Rho correlation coefficient, R; p value)	Correlation between SNP and resistance to levofloxacin (Spearman's Rho correlation coefficient, R; p value)	Correlation between SNP and resistance to gentamicin (Spearman's Rho correlation coefficient, R; p value)	Correlation between SNP and resistance to tobramycin (Spearman's Rho correlation coefficient, R; p value)
<i>mexG</i>	Ala58Thr	0.37; 0.06	0.67; < 0.01	0.81; < 0.01	0.81; < 0.01
<i>mexH</i>	Asp302Glu	0.22; 0.21	0.33; 0.09	0.66; < 0.01	0.66; < 0.01
<i>mexI</i>	Ala782Glu	0.09; 0.60	0.32; 0.11	0.53; < 0.01	0.53; < 0.01
<i>mexJ</i>	Phe13Leu	0.13; 0.50	0.42; 0.03	0.61; < 0.01	0.61; < 0.01
	Ala314Pro	0.03; 0.88	0.21; 0.29	0.57; < 0.01	0.40; 0.04
<i>mexK</i>	Lys694Arg	0.37; 0.06	0.32; 0.11	0.66; < 0.01	0.49; < 0.01
<i>mexL</i>	Ser4Pro	−0.11; 0.58	0.22; 0.28	0.45; 0.02	0.45; 0.02
<i>mexQ</i>	Pro172Leu	0.37; 0.06	0.67; < 0.01	0.81; < 0.01	0.81; < 0.01
	Gly505Asp	−0.16; 0.42	0.11; 0.57	0.37; 0.06	0.37; 0.06
<i>mexV</i>	Ala229Gly	0.11; 0.58	0.29; 0.14	0.25; 0.21	0.25; 0.21
<i>mexW</i>	Gly932Glu	0.37; 0.06	0.67; < 0.01	0.81; < 0.01	0.81; < 0.01
<i>mexX</i>	Ala30Thr	0.37; 0.06	0.50; 0.01	0.84; < 0.01	0.66; < 0.01
<i>mexY</i>	Ile536Val	0.37; 0.06	0.33; 0.09	0.66; < 0.01	0.66; < 0.01
	Gly589Ala	0.28; 0.16	0.48; 0.01	0.81; < 0.01	0.81; < 0.01
	Gln840Glu	0.01; 0.95	0.37; 0.06	0.54; < 0.01	0.71; < 0.01
	Asn1036Thr	0.22; 0.28	0.33; 0.09	0.66; < 0.01	0.66; < 0.01
<i>mexZ</i>	Gly89Ser	0.37; 0.06	0.67; < 0.01	0.81; < 0.01	0.81; < 0.01
<i>opmD</i>	Ala85Thr	0.37; 0.06	0.67; < 0.01	0.81; < 0.01	0.81; < 0.01
	Gly269Ser	0.22; 0.27	0.33; 0.09	0.66; < 0.01	0.66; < 0.01
	Ala270Gly	0.32; 0.10	0.21; 0.29	0.54; < 0.01	0.54; < 0.01
<i>opmE</i>	Ser46Gly	0.01; 0.95	−0.27; 0.18	−0.49; < 0.01	−0.49; < 0.01
	Pro59Ser	0.37; 0.06	0.47; 0.01	0.81; < 0.01	0.59; < 0.01
<i>parR</i>	Leu153Arg	0.27; 0.17	0.43; 0.02	0.60; < 0.01	0.42; 0.03
	Ser170Asn	0.32; 0.09	0.37; 0.06	0.71; < 0.01	0.54; < 0.01
<i>pmpM</i>	Phe422Ile	0.28; 0.16	0.48; 0.01	0.81; < 0.01	0.81; < 0.01
	Glu470Ala	0.27; 0.17	0.27; 0.17	0.60; < 0.01	0.60; < 0.01
<i>triA</i>	Gly34Ser	0.05; 0.82	0.22; 0.28	0.45; 0.02	0.45; 0.02
	Val304Asp	0.41; 0.03	0.76; < 0.01	0.90; < 0.01	0.70; < 0.01
	Gln316Glu	0.33; 0.09	0.57; < 0.01	0.73; < 0.01	0.54; < 0.01
	Gly317Ser	0.27; 0.17	0.43; 0.02	0.60; < 0.01	0.42; 0.03
	Ala319Val	0.41; 0.03	0.76; < 0.01	0.90; < 0.01	0.70; < 0.01

Italicised numbers indicate $p > 0.05$, **bold** numbers indicate $p < 0.05$.**Table 10**Correlation between SMR disinfectant resistance genes and antibiotics resistance of 16 *exoU* and 24 *exoS* keratitis isolates.

Antibiotics	Correlation between <i>qacEΔ1</i> and resistance to antibiotics (Spearman's Rho correlation coefficient, R; p value)	Correlation between <i>sugE1</i> and resistance to antibiotics (Spearman's Rho correlation coefficient, R; p value)
Ciprofloxacin	R = 0.46; $p < 0.01$	R = 0.30; $p > 0.05$
Levofloxacin	R = 0.67; $p < 0.01$	R = 0.69; $p < 0.01$
Gentamicin	R = 0.91; $p < 0.01$	R = 0.76; $p < 0.01$
Tobramycin	R = 1; $p < 0.01$	R = 0.67; $p < 0.01$

gentamicin resistance ($R = 0.91$, $p < 0.01$) and perfectly correlated with tobramycin resistance ($R = 1$, $p < 0.01$). For *sugE1*, there were a significant strong positive correlation with levofloxacin ($R = 0.69$, $p < 0.01$), gentamicin ($R = 0.76$, $p < 0.01$) and tobramycin ($R = 0.67$, $p < 0.01$) resistance.

4. Discussion

This study was undertaken to determine the differences in susceptibility of the two main lineages of *P. aeruginosa*, *exoU* and *exoS*, to four different MPDS. All MPDS killed all strains at 100 %. This is not surprising as all MPDS need to meet the International Organization for Standardization (ISO) Standard 14729, which states that multipurpose disinfecting solutions need to reduce the numbers of *P. aeruginosa* strain ATCC 9027 by $\geq 3 \log_{10}$ colony forming units. However, if contact lens wearers do not follow the advice of MPDS manufacturers they may inadvertently dilute their MPDS by, for example, reusing MPDS in cases and topping-up (topping-off) with new MPDS. This may then lead to loss of disinfecting activity. Even when OPTI-FREE Puremoist was diluted, it was the most effective MPDS for strains possessing *exoU*. This is similar to a previous study where this MPDS had highest activity against *Stenotrophomonas maltophilia*, also isolated from contact lens-related MK [34]. Conversely, Renu Advanced Formula was the most effective against *exoS* isolates.

The *exoU* isolates showed higher resistance against the tested antibiotics than *exoS*. This result aligns with the results obtained by previous

studies where *exoU* showed higher resistance to the fluoroquinolone and aminoglycoside antibiotics than *exoS* [28,30,45,46]. This study then went on to determine whether underlying genetic differences in strains were associated with susceptibility to the MPDS and antibiotics.

Different disinfectant resistance genes retrieved from the CARD database, and SNPs in those genes, provided novel insights into the genetic determinants of disinfectant and antibiotic resistance in *P. aeruginosa* keratitis isolates. The Majority of the significant mutations were more common in the *exoU* group compared to the *exoS* group, except for the Ser47Gly substitution in *opmE*. This aligns with previous work [23], which reported that *exoU* strains harboured a greater number of mutations in antibiotic resistance genes. The current findings extend this association to disinfectant resistance genes, suggesting that *exoU*-carrying strains may possess a broader genetic propensity for acquiring resistance mutations, potentially because of shared evolutionary pressures or compensatory mechanisms linking virulence and resistance.

The SNP in *mexQ* that led to a change in amino acid Arg1036His was associated with reduced MICs and MBCs to all MPDS. *MexQ* is a part of the multidrug efflux pump MexPQ-OpmE which is associated with resistance to several antibiotics as well as disinfectants (<https://card.mcmaster.ca/ontology/40351>). This result suggests this particular SNP may be associated with a defect in the pump's activity. On the other hand, the genetic change leading to Gly505Asp change in *MexQ* was associated with increased resistance to most of the MPDS, suggesting this may activate the pump. A similar hypothesis can be mounted for the SNPs in *mexP* and *opmE* that were associated with increased resistance. These hypotheses require further investigation. *OpmD* is a part of the efflux pump MexGHI-OpmD that transports phenazine, which in turn controls biofilm formation, and transports antibiotics [47,48]. *OmpD* has been reported previously to be associated with increase antibiotic resistance [49], and in the current study, the SNP resulting in the Gly269Ser substitution in *ompD* was associated with increased MICs and MBCs to most of the MPDSs. Several amino acid changes in *triA* were moderately associated with MPDS resistance and the *triABC-OmpH* efflux pump has previously been reported to efflux out the disinfectant triclosan [50]. Other SNPs were found, more sporadically, associated with resistance to the MPDS, and further work is required to definitively show causal links.

The prevalence of different SMR disinfectant resistance genes varied between the two *exo* groups. Almost all isolates (except PA176 from the *exoS* group) harboured the *qacE* gene. This confirms a previous report with as subset of the strains in the current investigation [43]. However, in another study from non-keratitis clinical strains of *P. aeruginosa*, only 23.8 % (15 out of 63) strains carried this gene [51]. This previous study [51] was from 1998, and it may be that strains have acquired *qacE* since that time. The presence of *sugE1* and *qacEΔ1* differed significantly, with a higher frequency in the *exoU* strains than the *exoS* strains. Overall, 37.5 % *P. aeruginosa* possessed *qacEΔ1*. This result is comparable with previous studies where between 17.5 % [44] and 65.1 % [51] of *P. aeruginosa* isolates possessed this gene. In the current study, the overall presence of the *sugE1* in *P. aeruginosa* was 48 % which is lower than a previous report where 95 % *P. aeruginosa* strains carried *sugE1* gene [44]. Previous studies have found no association between possession of *qacE* or *qacEΔ1* and quaternary ammonium compounds resistance in *P. aeruginosa* [43,52]. In contrast, possession of disinfectant resistance genes in *Staphylococcus aureus* resulted in more resistance to the disinfectants than those isolates lacking the genes [53]. The current study was able to identify correlations between possession of *qacEΔ1* or *sugE1* and MICs and MBCs to Renu Advanced Formulation, Biotrue and Acuvue RevitaLens. For Renu Advanced Formulation, the possession of either gene explained approximately 25 % to 33 % of the variation in MICs or MBCs to this MPDS, and for Biotrue between 18 % and 29 % of the variation in its MIC. Possession of *sugE1* explained 17 % of the variation in the MIC to Acuvue RevitaLens. *QacEΔ1* is found on integrons and has been found on a transposon in *P. aeruginosa* PA34 [54].

This suggests it is highly mobile. This gene and *sugE1* encode efflux pump proteins [12,55] and so this is likely to be the mechanism of resistance.

Association of SNPs in the CARD derived disinfectant genes, and the possession of SMR disinfectant resistance genes (*qacE*, *qacEΔ1* and *sugE1*), with antibiotics resistance was also determined in the current study. SNPs in the efflux pumps components and regulatory genes (*mexG*, *mexJ*, *mexQ*, *mexW*, *mexX*, *mexY*, *opmD*, *opmE*, *triA*, *pmpM*, *mexZ*, and *parR*) showed associations with levofloxacin resistance and all the significantly different mutations (except Gly505Asp *mexQ* and Ala229Gly *mexV*) showed association with both gentamicin and tobramycin resistance. Association of efflux pumps *mexGHI-OpmD*, *mexJKI-OprM*, *mexPQ-OprE*, *mexVW-OprM*, *mexXY-OprM* and mutations in the efflux pumps regulatory genes (*mexZ*, *parR*) with resistance to different groups of antimicrobial agents including fluoroquinolones and aminoglycosides antibiotics have been reported in several other studies [18,48,56–60]. Whilst all the significantly differently prevalent novel mutations between *exoU* and *exoS* (except Gly89Ser *mexZ*) did not show any presumed effect on functional changes of the proteins, the novel nature of all reported mutations warrants the need for functional validation to confirm their causal roles in resistance. While prior studies have focused on established resistance markers, the current work identifies previously unrecognized SNPs that may contribute to emerging resistance phenotypes in ocular infections. These results highlight potential overlaps between disinfectant and antibiotic resistance mechanisms, possibly mediated by shared efflux systems or stress response pathways (e.g. the *parRS* two component system).

In case of SMR disinfectant resistance genes, as presence of the *qacE* gene was detected in all *exoU* and *exoS* (except one isolate) strains, no correlations was observed between this gene and antibiotic resistance, which is supported by a previous study [44]. In case of *qacEΔ1*, a large positive correlation was found with resistance to levofloxacin, gentamicin and tobramycin. A previously conducted study reported that 76.2 % *qacEΔ1* carrying *P. aeruginosa* were MDR [44]. Another study, performed with environmental *P. aeruginosa*, and found that *qacEΔ1* positive strains were resistant to different groups of antibiotics including aminoglycosides [51]. Similar findings have also been reported by other studies [61,62]. Not only can *qacEΔ1* produce an efflux pump, but its association with class 1 integrons and plasmids in Gram-negative bacteria is likely to be another reason for cross-resistance to antibiotics as these can also be transfer between strains by these mechanisms [63–66].

Another disinfectant resistance *sugE1* gene showed no significant correlation with ciprofloxacin but was strongly associated with levofloxacin and both aminoglycosides. This is in contrast with the finding of a previous study, where there was no correlation with *sugE1* gene and antibiotics resistance [44]. *SugE1* can be associated with plasmids of *Enterobacteriaceae* including *E. coli*, *Salmonella* spp., *Shigella* spp., *Klebsiella* spp. [67–71]. In *E. coli*, overexpression of *sugE1* is associated with resistance to a narrow range of quaternary ammonium compounds and cationic dyes [72]. However, in *P. aeruginosa*, the role of *SUG* proteins in antibiotics resistance has not been well studied which requires further investigation with a larger sample size.

5. Conclusion

Strains of *P. aeruginosa* from MK carrying *exoU* were more likely to be resistant to the tested MPDS and antibiotics than the *exoS* strains. The frequency of many SNPs in the disinfectant resistance genes were significantly higher in *exoU* group compared to *exoS*, and the presence of these disinfectant resistance genes and their SNPs was correlated with phenotypic susceptibility to MPDS. Associations were also found between antibiotic resistance and possession of SMR disinfectant resistance genes (*qacEΔ1* and *sugE1*) or SNPs in other disinfectant-associated resistance genes.

The current study used only four MPDS. Therefore, future experiments with other MPDS and larger numbers of *exoU* and *exoS* isolates

from both contact lens and non-contact lens might strengthen the significant correlations found between disinfectant resistance genes with phenotypic resistance to MPDS. In addition, novel mutations were observed in our study which need follow-up hypothesis-driven studies to understand their effect on resistance to components of the MPDS and fluoroquinolones and aminoglycosides.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement

Data is provided within the manuscript or [supplementary information files](#).

Ethics approval

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clae.2025.102436>.

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